NeuroScan AI - Feature Inventory

This document provides a comprehensive list of all major and minor features implemented in the NeuroScan AI project.

🚀 1. Core AI & Diagnostics

Big Features:

- **Multi-Model Consensus Engine:** Combines **EfficientNetV2** (Primary), **Swin Transformer V2**, and **CoAtNet-2** to provide a weighted diagnostic opinion.

- **Tumor Localization & Segmentation:** Automatically detects tumor regions (ROI) and calculates bounding boxes using Gradient Class Activation Mapping (Grad-CAM) and intensity thresholding.

- **Real-time Clinical Metrics:**

  - **Morphology Analysis:** Distinguishes between smooth (Round/Ovoid) and irregular (Spiculated) tumor shapes.

  - **Dimension Estimation:** Auto-calculates tumor width, height, and depth in centimeters.

  - **Lobe Localization:** Identifies the brain region (e.g., Frontal Lobe, Temporal Lobe) where the anomaly sits.

  - **Tissue Characterization:** Detects Necrosis (dead tissue) and Vascularity (blood flow anomalies).

- **Urgency Classification:** Auto-assigns "ROUTINE", "STAT", or "CRITICAL" priority based on tumor size and confidence.

Small Features:

- **OOD (Out-of-Distribution) Filter:** Rejects low-quality or non-MRI images before analysis to prevent hallucinations.

- **Visual "Bright Mass" Check:** Heuristic fallback to detect hyper-intense tumors even if the ML model is uncertain.

- **Confidence Calibration:** Adjusts raw probabilities to prevent over-confident "No Tumor" results on complex cases.

🧠 2. Advanced Visualization Suite

Big Features:

- **Holographic 3D Brain Viewer:** Interactive webGL (Three.js) model showing the brain structure with a floating, glowing "Tumor Sphere" located at the precise coordinates of the detection.

- **2D "Blueprint" Mode:** A stylized, high-contrast schematic view of the scan for structural analysis.

- **Thermal Vision Mode:** Pseudocolor heatmap overlay to highlight intensity hotspots (simulating thermal activity).

- **Pixel Probe (HU):** Interactive tool to inspect specific pixel intensity values (simulating Hounsfield Units).

Small Features:

- **Dynamic Particles:** "Glitch" particles and connecting neurile lines in the 3D view that react to tumor presence.

- **Medical Image Adjustment:** Real-time Brightness and Contrast sliders (Window/Leveling).

- **Magnifying Lens:** Localized zoom tool for inspecting fine details.

- **Symmetry Grid:** Overlay grid to compare left/right brain hemisphere symmetry.

- **Edge Detection filter:** Sobel operator visualization to highlight structural boundaries.

🏥 3. Tele-Radiology & Collaboration

Big Features:

- **Tele-Consultation Portal:** A fully functional interface for peer-to-peer video calls (simulated UI with camera/mic controls).

- **Encrypted Chat System:** Real-time messaging interface with "E2E Encrypted" status indicators.

- **QR Code Mobile Sync:** Generates a functional QR code to "handoff" the session to a mobile device.

- **"Dr. Furqan" AI Persona:** A voice-activated AI assistant that can answer questions about the scan verbally (Text-to-Speech) and listen to user queries (Speech-to-Text).

Small Features:

- **Live Audio Waveform:** Real-time visualizer that reacts to the AI's voice.

- **Emergency Desk Speed Dial:** Quick-access button to contact emergency services (simulated).

- **Screen Sharing UI:** Controls for sharing the diagnostic view with remote peers.

- **Latency Monitoring:** Simulated network stats (Ping, Jitter, Bitrate) for the telehealth connection.

⚡ 4. Workflow & Productivity

Big Features:

- **Automated PDF Reports:** Generates professional, downloadable PDF reports containing the scan image, key findings, and AI analysis.

- **Smart Scheduler:** Recommends follow-up appointment intervals (e.g., "3 Months" vs "1 Year") based on diagnosis urgency.

- **Clinical Protocol Checklist:** Interactive checklist for standard procedures (MRI, Labs, Referral) that tracks completion.

- **AI Note Generator:** Auto-drafts a full radiology report text (History, Findings, Impression) which can be read aloud.

Small Features:

- **Voice Dictation:** Microphone buttons on "Symptoms" and "Notes" fields to input text via voice.

- **Secure Share Link:** Generates a unique, time-limited link to share the report with patients.

- **Blind Mode:** Privacy feature that blurs patient PII (Name, Age, ID) on the dashboard for HIPAA compliance during demos.

- **History Dashboard:** Timeline chart showing tumor vs. non-tumor detection trends over the last 7 days.

🌐 5. Futuristic & Experimental

Big Features:

- **Federated Learning Node:** A simulation of a decentralized training network, complete with a "Global Swarm" visualization and local training logs.

- **Token Rewards ($NEURO):** Gamified "Contribution Reward" tracker showing earned tokens for participating in federated learning.

Small Features:

- **Terminal Logs:** Scrolling command-line interface showing "real-time" connection events and model updates.

- **GPU Simulation:** UI indicators showing "NVIDIA A100" usage and hardware acceleration status.

🎨 6. UI/UX & Aesthetics

Big Features:

- **Glassmorphism Design:** Premium, translucent UI elements with background blur (`backdrop-filter`) and border gradients.

- **Responsive Mobile View:** Fully optimized layout for tablets and smartphones.

- **Internationalization (i18n):** Support for 7 languages (English, Spanish, French, German, Hindi, Telugu, Chinese).

Small Features:

- **Micro-Animations:** Hover effects, pulse animations on status dots, and smooth transitions between pages.

- **Dynamic Greetings:** "Welcome back" messages and context-aware feedback.

- **Custom Scrollbars:** Sleek, thin scrollbars that match the dark theme.

- **Loading Skeletons & Spinners:** Polished loading states for all async actions.